

[Quiz—1] Spring 2024: CSE 309: Theory of Automata

— SOLUTION —

Thursday, February 1, 2024. Total Marks: 10, Duration: 15 minutes

Question 1

[10 points]

For each of the following languages, give two strings that are members of the language (if any) and two strings which are not members (if any). Assume that the alphabet, $\Sigma = \{0, 1\}$.

(a) $(0101)^*$

Solution:

Members: $\{\epsilon, 0101, 01010101, 010101010101, \dots\}$

Non-members: $\{0, 1, 010, 101, 01010, 10101, \dots\}$

(b) $(0 \cup 01 \cup 11)\Sigma^*$

Solution:

Members: $\{0, 01, 11, 00, 010, 110, 111, 000, 001, 010, 011, 110, 111, \dots\}$

Non-members: $\{\epsilon, 1, 10, 101, 100, 1010, 1011, 1100, 1101, 1110, 1111, \dots\}$

(c) $(0 \cup 01)\Sigma^*\emptyset$

Solution:

Members: $\{\}$

Non-members: $\{\epsilon, 0, 1, 10, 11, 100, 101, 110, 111, 1000, 1001, 1010, 1011, \dots\}$